

Frequency-Domain Locking of a Pulsed Microring Single-Photon Source via Low-Frequency Pump Transmission Extraction

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Abstract Here, we report frequency-domain locking of a pulsed microring single-photon source by extracting the transmission spectrum derivative via low-frequency modulation. The generated zero-crossing error signal effectively compensates for resonance drift, enabling long-term, highly stable system operation. ©2026 The Author(s)

Introduction

Integrated microring resonators (MRRs) have emerged as a premier platform for constructing scalable quantum light sources via spontaneous four-wave mixing (SFWM)[1, 2]. Compared to continuous-wave (CW) operation, pulsed pumping offers significant advantages for practical quantum information processing. It provides a deterministic timing reference essential for establishing synchronous quantum logic links and significantly enhances the coincidence-to-accidental ratio (CAR). However, the extremely high quality factor (Q) of MRRs renders their resonance wavelength highly susceptible to environmental thermal fluctuations and pump-induced thermal nonlinearities[3, 4]. Maintaining precise and long-term spectral alignment between the pulsed pump light and the cavity resonance remains a critical bottleneck. Conventional stabilization techniques typically rely on direct current (DC) power detection[5, 6]. Such schemes suffer from "directional blindness" at the resonance extremum and are inherently vulnerable to laser intensity fluctuations[7]. Furthermore, existing closed-loop control systems for high-repetition-rate (e.g., 500 MHz) pulsed signals often require expensive, high-bandwidth analog-to-digital converters (ADCs) and complex microwave electronics, which severely hinders scalable chip-scale integration[8, 9].

In this work, we propose and experimentally demonstrate a robust frequency-domain locking architecture tailored for pulsed microring single-photon sources. Specifically, by applying a minute, low-frequency sinusoidal dither to the integrated micro-heater, we deliberately induce a periodic perturbation in the cavity resonance. This mechanism allows us to shift the control paradigm from conventional absolute power tracking to the direct extrac-

tion of the transmission spectrum's first derivative. Utilizing phase-sensitive coherent demodulation, this dynamic spectral probing isolates a bipolar error signal that exhibits a strict zero-crossing precisely at the resonance extremum. Consequently, the detuning direction is unambiguously identified, completely decoupling the resonance stabilization process from transient pump intensity fluctuations and fiber-to-chip coupling variations. By feeding this derivative-based error signal into a closed-loop PID feedback system, continuous and precise wavelength matching is achieved. Crucially, our approach entirely circumvents the necessity for high-speed ADC sampling and complex microwave electronics, thereby ensuring long-term, highly stable system operation while offering a highly scalable, chip-compatible solution for integrated quantum photonic networks.

Experimental Setup and Principle

The experimental configuration for the self-stabilized pulsed microring single-photon source is illustrated in Figure 1(a). A pulsed laser, featuring a 500 MHz repetition rate and a 30 ps pulse width, is employed to pump the MRR. The proposed integrated MRR quantum photonic source is fabricated on a passive silicon-on-insulator (SOI) platform with standard 450 nm × 220 nm waveguides. To actively stabilize the resonance wavelength against thermal drifts, a 10 kHz low-frequency sinusoidal dither voltage is applied to the integrated micro-heater. At the through-port of the MRR, a dense wavelength-division multiplexer (DWDM) separates the generated single photons from the strong residual pump light. The residual pump is then detected by a photodetector (PD) and amplified by a transimpedance amplifier (TIA). This electrical signal is subsequently fed into a phase-

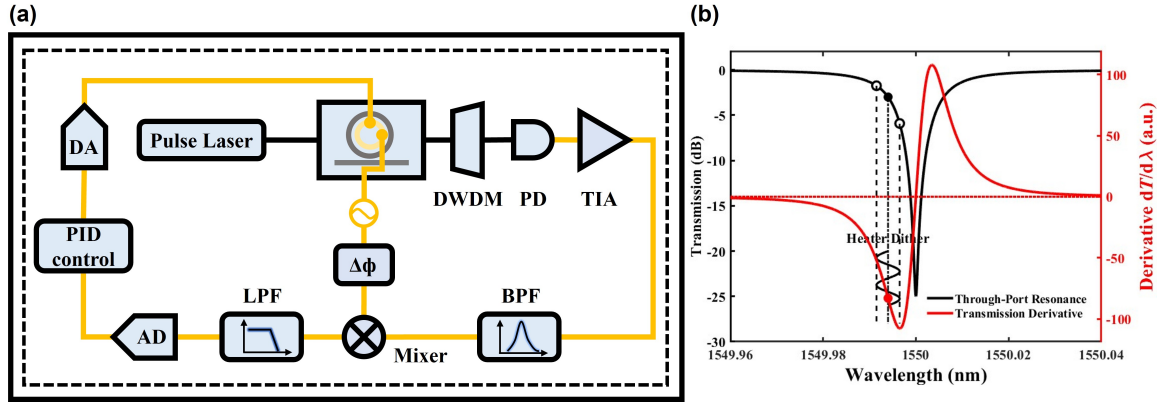


Fig. 1: Derivative-based frequency locking architecture for pulsed microring single-photon sources. (a) Experimental setup. A low-frequency dither is applied to the micro-heater, and the error signal is extracted from the residual pump light via phase-sensitive detection. A digital PID controller closes the feedback loop. (b) Locking principle. The extracted fundamental AC component (red curve) is strictly proportional to the first derivative of the optical transmission (black curve). This provides a robust, zero-crossing error signal at the resonance extremum, which is completely decoupled from pump intensity fluctuations. (DWDM: dense wavelength-division multiplexer; PD: photodetector; TIA: transimpedance amplifier; BPF/LPF: bandpass/low-pass filter).

sensitive detection loop—comprising a bandpass filter (BPF), a mixer, and a low-pass filter (LPF)—to extract a stable error signal. This analog error voltage is then digitized by a high-speed analog-to-digital converter (ADC, AD9248). An FPGA-based digital PID controller processes this signal to calculate the compensation value, which is subsequently converted back to an analog voltage via a digital-to-analog converter (DAC, AD9767) to drive the micro-heater, thereby closing the feedback loop.

This frequency-domain locking scheme is fundamentally based on extracting the first derivative of the optical transmission spectrum via lock-in amplification. By applying a minute low-frequency dither $\delta(t) = \bar{\delta} + \delta_m \sin(\Omega t)$ (where the modulation frequency $\Omega/2\pi = 10$ kHz) to the micro-heater, the instantaneous optical transmission $T(\delta(t))$ undergoes a periodic modulation. Based on a second-order Taylor expansion around the static detuning $\bar{\delta}$, the transmission can be expressed as:

$$T(\delta(t)) \approx T(\bar{\delta}) + T'(\bar{\delta})\delta_m \sin(\Omega t) + \frac{1}{2}T''(\bar{\delta})\delta_m^2 \sin^2(\Omega t)$$

After optoelectric conversion, the BPF isolates the fundamental alternating-current (AC) component at frequency Ω , effectively discarding the direct-current (DC) and higher-order harmonic terms. The filtered electrical signal is strictly proportional to the first derivative of the transmission spectrum:

$$V_{AC}(t) \propto T'(\bar{\delta})\delta_m \sin(\Omega t + \phi_{sys})$$

where ϕ_{sys} represents the system phase delay. This fundamental AC component is then multiplied by a synchronous reference wave (e.g., $\sin(\Omega t + \phi_{comp})$) in the mixer. Applying trigonometric product-to-sum identities, the mixed signal becomes:

$$V_{mix}(t) \propto T'(\bar{\delta}) [\cos(\phi_{sys} - \phi_{comp}) - \cos(2\Omega t + \phi)]$$

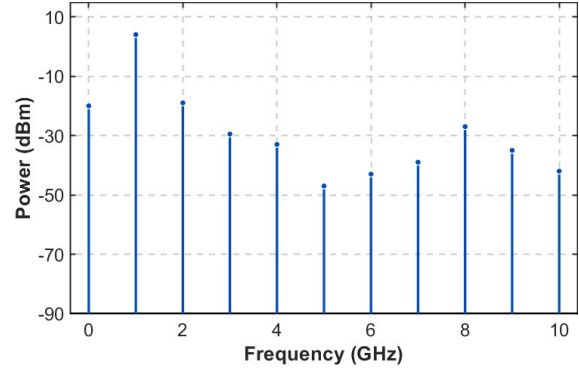


Fig. 2: Radio-frequency spectrum of the pulsed pump signal. The discrete spectral lines exhibit a strict 500 MHz spacing, corresponding to the fundamental repetition rate of the mode-locked laser.

After time integration via the LPF, the 2Ω high-frequency alternating term is entirely eliminated. The ultimate pure DC error signal $\varepsilon(\bar{\delta})$ can be expressed as:

$$\varepsilon(\bar{\delta}) \propto T'(\bar{\delta}) \cos(\Delta\phi)$$

where $\Delta\phi = \phi_{sys} - \phi_{comp}$ is the residual phase difference between the system signal and the reference square wave. When optimally phase-compensated ($\Delta\phi = 0$), the extracted DC error voltage is directly proportional to the transmission derivative. As depicted in Fig. 1(b), this mechanism provides a robust, bipolar zero-crossing signature precisely at the resonance extremum, which unambiguously indicates the detuning direction and fundamentally decouples the resonance stabilization from absolute pump power fluctuations.

To fully characterize the driving pump source, the radio-frequency (RF) spectrum of the pulsed laser was measured, as presented in Fig. 2. The spectrum exhibits a characteristic discrete comb-like structure with adjacent spectral lines separated by a strict 500 MHz spacing, which precisely corresponds to the fundamental repetition rate of

the mode-locked laser.

To systematically evaluate the efficacy of the proposed frequency-domain locking mechanism, we characterized the coincidence count rate (CCR) and the CAR of the pulsed microring single-photon source as a function of pump power, as depicted in Fig. 3. Fig. 3(a) illustrates the CCR across a pump power sweeping range of 3 dBm to 13 dBm. Under active stabilization (locked state), the system precisely maintains optimal resonance. Consequently, the generation rate exhibits a rigorous quadratic dependence on the pump power, confirming the underlying spontaneous four-wave mixing (SFWM) process. In stark contrast, the free-running (unlocked) state suffers from severe count rate degradation, especially in the high-power regime where elevated intracavity circulating power exacerbates thermal-induced detuning.

The quantum purity of the emitted photon pairs is quantified by the CAR, as shown in Fig. 3(b). Both configurations display the characteristic power-dependent CAR evolution: the ratio is bounded by detector dark counts at the low-power threshold and deteriorates at high pump levels due to the escalating probability of multi-photon pair emission. The optimal CAR is observed at a pump power of approximately 6 dBm. Benefiting from the robust suppression of thermal drifts, the locked source achieves a peak CAR of nearly 230, substantially outperforming its free-running counterpart (peak CAR $\approx$ 190). More importantly, the frequency-domain locking architecture sustains a markedly enhanced CAR across the entire scanning range, underscoring its indispensability for the stable generation of high-purity quantum light.

Conclusions

To overcome thermal instabilities in silicon MRRs under pulsed pumping, we demonstrated a robust frequency-domain locking architecture. By applying a low-frequency dither, the system dynamically extracts the first derivative of the pump transmission to generate a zero-crossing error signal. This mechanism fundamentally decouples resonance stabilization from absolute pump intensity and coupling fluctuations. Experimentally, the closed-loop system achieved drift-free locking, strictly preserving the quadratic scaling of the CCR. Furthermore, by effectively suppressing thermal drifts, the locked source elevated the peak CAR to nearly 230, significantly outperforming the free-running state (CAR $\approx$ 190). By circumventing complex high-speed microwave electronics, this derivative-based stabilization scheme provides a highly scalable, chip-compatible architectural foundation for large-scale integrated quantum photonic networks.

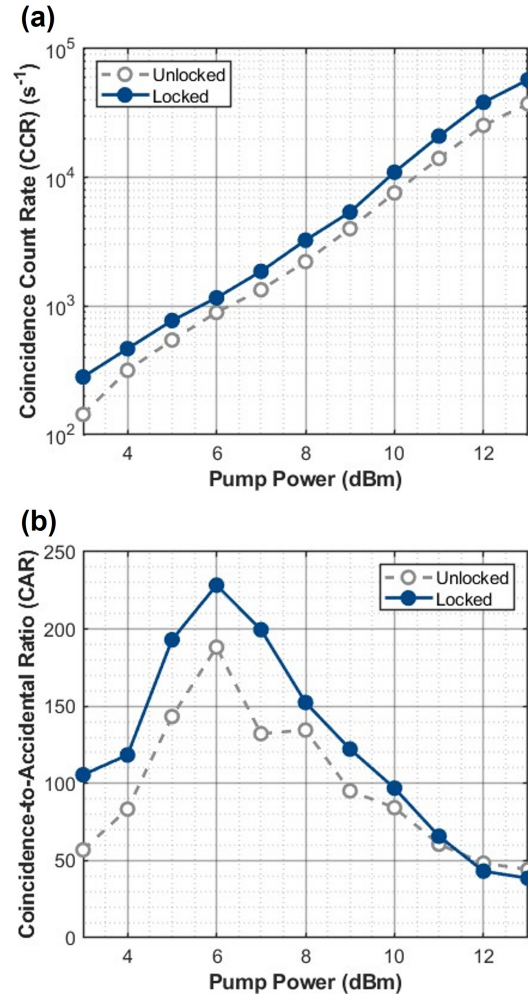


Fig. 3: Performance of the pulsed microring single-photon source. (a) Coincidence count rate (CCR) versus pump power. The locked state (solid blue) maintains a robust quadratic scaling, while the unlocked state (dashed gray) degrades due to thermal detuning. (b) Coincidence-to-accidental ratio (CAR) versus pump power. Frequency-domain locking effectively suppresses thermal drifts, preserving a significantly higher CAR across the entire sweeping range.

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References

- [1] K. Delfanazari, "Chip-scale electrically driven superconducting coherent photon sources for quantum information processing", *Nature Photonics*, vol. 19, no. 11, pp. 1163–1177, 2025. DOI: [10.1038/s41566-025-01735-9](https://doi.org/10.1038/s41566-025-01735-9).
- [2] B. Hou, D. Chen, H. Ma, Z. Zhao, Q. Liang, B. Zhang, L. Ye, Q. Huang, D. Wang, T. Dai, H. Zhu, Y. Wang, J. Yang., "Chip-scale high spectral purity micro-ring photon-pair source based on dual-pulse control", *Optics Express*, vol. 34, no. 7, pp. 11 742–11 751, 2026. DOI: [10.1364/OE.589347](https://doi.org/10.1364/OE.589347).
- [3] H. Jayatileka, H. Shoman, L. Chrostowski, S. Shekhar, "Photoconductive heaters enable control of large-scale silicon photonic ring resonator circuits", *Optica*, vol. 6, no. 1, pp. 84–91, 2019. DOI: [10.1364/OPTICA.6.000084](https://doi.org/10.1364/OPTICA.6.000084).
- [4] Q. Zhu, C. Qiu, Y. He, Y. Zhang, Y. Su, "Self-homodyne wavelength locking of a silicon microring resonator", *Optics Express*, vol. 27, no. 25, pp. 36 625–36 636, 2019. DOI: [10.1364/oe.27.036625](https://doi.org/10.1364/oe.27.036625).
- [5] H. Liu, T. Dai, Y. Wang, C. Wang, B. Hou, Y. Wang, J. Yang, "Automatic wavelength locking of microring resonators based on coherent detection", in *2024 Asia Communications and Photonics Conference (ACP) and International Conference on Information Photonics and Optical Communications (IPOC)*, IEEE, 2024, pp. 1–4. DOI: [10.1109/ACP/IPOC63121.2024.10809694](https://doi.org/10.1109/ACP/IPOC63121.2024.10809694).
- [6] K. Padmaraju, D. F. Logan, T. Shiraishi, J. J. Ackert, A. P. Knights, K. Bergman, "Wavelength locking and thermally stabilizing microring resonators using dithering signals", *Journal of Lightwave Technology*, vol. 32, no. 3, pp. 505–512, 2013. DOI: [10.1109/JLT.2013.2294564](https://doi.org/10.1109/JLT.2013.2294564).
- [7] B. Hou, H. Ma, Z. Zhao, Q. Liang, F. Ruan, Q. Huang, D. Chen, B. Zhang, T. Dai, Y. Wang, H. Zhu, J. Yang., "Silicon bidirectional-pumping-assisted multiplexing single-photon sources with high count rates", *Optics Letters*, vol. 51, no. 6, pp. 1661–1664, 2026. DOI: [10.1364/OL.586532](https://doi.org/10.1364/OL.586532).
- [8] J. M. Arrazola, V. Bergholm, K. Brádler, T. R. Bromley, M. J. Collins, I. Dhand, A. Fumagalli, T. Gerrits, A. Goussev, L. G. Helt, et al., "Quantum circuits with many photons on a programmable nanophotonic chip", *Nature*, vol. 591, no. 7848, pp. 54–60, 2021. DOI: [10.1038/s41586-021-03202-1](https://doi.org/10.1038/s41586-021-03202-1).
- [9] D. Mohtashemi, M. M. Green, "A low-power 8-gs/s comparator for high-speed analog-to-digital conversion in 0.13 μ m cmos technology", *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 66, no. 4, pp. 557–561, 2018. DOI: [10.1109/TCSII.2018.2869826](https://doi.org/10.1109/TCSII.2018.2869826).